# JEE–MAIN EXAMINATION – JANUARY 2026

**(HELD ON WEDNESDAY 28th JANUARY 2026)** **TIME : 9:00 AM TO 12:00 NOON**

## MATHEMATICS — TEST PAPER WITH ANSWER

### SECTION-A

**1.** If $g(x) = 3x^2 + 2x - 3, f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

(1) $\frac{25}{6}$ (2) $-\frac{25}{6}$

(3) $\frac{7}{2}$ (4) $-\frac{7}{2}$

**2.** The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{k(k+1)}{k!}\right)$ is :

(1) 2/e (2) 1/e

(3) $\sqrt{e}$ (4) e/2

**3.** Let $y = x$ be the equation of a chord of the circle $C_1$ (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let $C_2$ be another circle described on the given chord as its diameter. If the equation of the chord of the circle $C_2$, which passes through the point (2, 3) and is farthest from the center of $C_2$, is $x + ay + b = 0$, then $a - b$ is equal to :

(1) 10 (2) –6

(3) –2 (4) 6

**4.** If $\frac{\tan(A-B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1$, $A, B, C \in \left(0, \frac{\pi}{2}\right)$, then

(1) tan A, tan C, tan B are in G.P.

(2) tan A, tan B, tan C are in G.P.

(3) tan A, tan C, tan B are in A.P.

(4) tan A, tan B, tan C are in A.P.

**5.** Let z be a complex number such that $|z - 6| = 5$ and $|z + 2 - 6i| = 5$.

Then the value of $z^3 + 3z^2 - 15z + 141$ is equal to

(1) 42 (2) 37

(3) 50 (4) 61

**6.** Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x + 2\sqrt{2}y = 4$. If the co-ordinates of the vertex A are $(\alpha, \beta)$, then the greatest integer less than or equal to $|\alpha + \sqrt{2}\beta|$ is

(1) 2 (2) 3

(3) 5 (4) 4

**7.** Let S = {1, 2, 3, 4, 5, 6, 7, 8, 9} . Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

(1) 29x = 5y (2) 45x = 7y

(3) 21x = 4y (4) 56x = 9y

**8.** Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N \text{ and } a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S, which are divisible by $x^2 + 2$, is

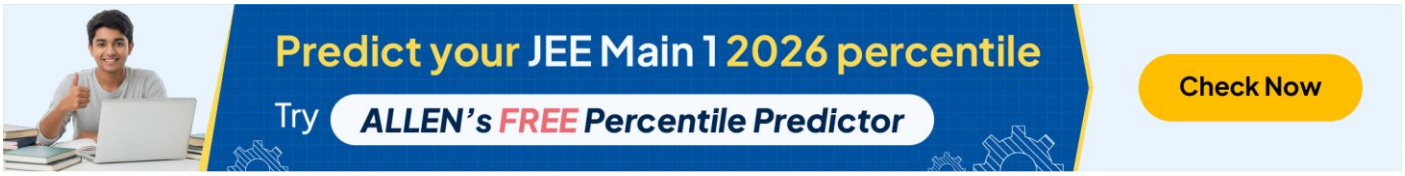
(1) 20 (2) 6

(3) 120 (4) 10

**9.** A bag contains 10 balls out of which k are red and (10 – k) are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains l red and 9 black balls is :

(1) $\frac{7}{11}$ (2) $\frac{7}{55}$

(3) $\frac{7}{110}$ (4) $\frac{14}{55}$



**10.** If $\alpha, \beta$, where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of $\lambda$ is :

(1) 6 (2) 2

(3) 4 (4) 8

**11.** If $\int\left(\frac{1-5\cos^2 x}{\sin^5 x \cos^2 x}\right)dx = f(x) + C$ where C is the constant of integration, then $f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right)$ is equal to

(1) $\frac{1}{\sqrt{3}}\left(26+\sqrt{3}\right)$ (2) $\frac{4}{\sqrt{3}}\left(8-\sqrt{6}\right)$

(3) $\frac{1}{\sqrt{3}}\left(26-\sqrt{3}\right)$ (4) $\frac{2}{\sqrt{3}}\left(4+\sqrt{6}\right)$

**12.** Let $f$ be a polynomial function such that $f(x^2 + 1) = x^4 + 5x^2 + 2$, for all $x \in \mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

(1) $\frac{41}{3}$ (2) $\frac{33}{2}$

(3) $\frac{27}{2}$ (4) $\frac{5}{3}$

**13.** The area of the region $R=\{(x, y): xy \le 8, 1 \le y \le x^2, x \ge 0\}$ is

(1) $\frac{1}{3}\left(49\log_e(2)-15\right)$

(2) $\frac{2}{3}\left(20\log_e(2)+9\right)$

(3) $\frac{2}{3}\left(24\log_e(2)-7\right)$

(4) $\frac{1}{3}\left(40\log_e(2)+27\right)$

**14.** The value of

$$\lim_{x\to 0}\frac{\log_e\left(\sec(ex).\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2 - e^{2\cos x}}$$

is equal to

(1) $\frac{(e^{10}-1)}{2e^2(e^2-1)}$ (2) $\frac{(e^{20}-1)}{2e^2(e^2-1)}$

(3) $\frac{(e^{20}-1)}{2(e^2-1)}$ (4) $\frac{(e^{10}-1)}{2(e^2-1)}$

**15.** The mean and variance of 10 observations are 9 and 34.2, respectively. If 8 of these observations are 2, 3, 5, 10, 11, 13, 15, 21, then the mean deviation about the median of all the 10 observations is

(1) 5 (2) 4

(3) 6 (4) 7

**16.** Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$. If $BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $CB\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is

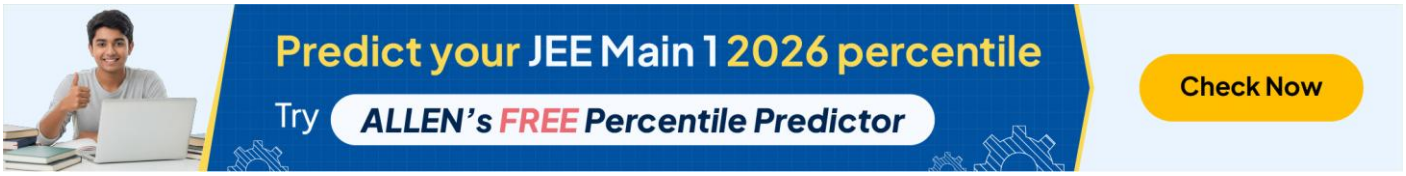
(1) 2 (2) 0

(3) –2 (4) 4

**17.** The common difference of the A.P.: $a_1, a_2, \ldots, a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots, b_n$. If $b_{31} = -277$, $b_{43} = -385$ and $a_{78} = 327$, then $a_1$ is equal to

(1) 21 (2) 24

(3) 19 (4) 16



**18.** If the distances of the point (1, 2, a) from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines

$L_1:\frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and

$L_2:\frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal,

then a + b + c is equal to

(1) 7 (2) 5

(3) 6 (4) 4

**19.** For three unit vectors $\vec{a},\vec{b},\vec{c}$ satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$ and

$|2\vec{a}+k\vec{b}+k\vec{c}|=3,$

the positive value of k is :

(1) 3 (2) 6

(3) 4 (4) 5

**20.** Let y = y(x) be the solution of the differential equation

$x\frac{dy}{dx}-\sin 2y=x^3(2-x^3)\cos^2 y,\ x\neq 0.$

If y (2) = x, then tan(y(1)) is equal to

(1) $\frac{3}{4}$ (2) $\frac{7}{4}$

(3) $-\frac{7}{4}$ (4) $-\frac{3}{4}$

## SECTION-B

**21.** In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R}-(a, b)$, then $a^2+b^2$ is equal to ____.

**22.** For some $\theta\in\left(0,\frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2-y^2\sec^2\theta=8$ be $e_1$ and $\ell_1$, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2\sec^2\theta+y^2=6$ be $e_2$ and $\ell_2$, respectively. If $e_1^2=e_2^2\left(\sec^2\theta+1\right)$, then $\left(\frac{\ell_1\ell_2}{e_1e_2}\right)\tan^2\theta$ is equal to _____.

**23.** If $k=\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right)+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$ then the number of solutions of the equation $\sin^{-1}(kx-1)=\sin^{-1}x-\cos^{-1}x$ is _____.

**24.** The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x\,|\sin\pi x|\,dx\right)}\right|\right)$ is ____.

**25.** Let PQR be a triangle such that $\overrightarrow{PQ}=-2\hat{i}-\hat{j}+2\hat{k}$ and $\overrightarrow{PR}=a\hat{i}+b\hat{j}-4\hat{k}$, a, b $\in\mathbb{Z}$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $|\overrightarrow{PR}|=9$ and $\overrightarrow{PS}=\hat{i}-7\hat{j}+2\hat{k}$, then the value of 3a – 4b is ___.

**Allen**

**NTA**



ALLEN

# For Class 12th Pass Students

## RISE. REPEAT.
## RANK UP IN JEE




## JOIN LEADER COURSE
**JEE (Main+Adv.) 2027**

26th Mar & 15th Apr

Know more

ALLEN ONLINE

# Think JEE 2027 will be your best shot?

Join the **Leader Online Course!**

Win up to **90%** scholarship via **ASAT**

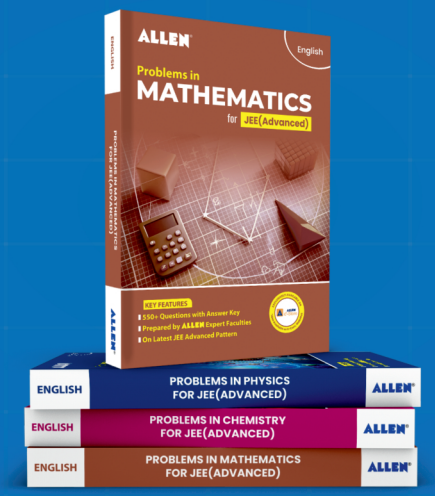


Enrol Now

# DON'T PREPARE HARDER
# PREPARE RIGHT !

Advanced-level MCQs

Prepared By ALLEN Experts




Well Graded Exercise

Detailed Answer Key

SHOP NOW on ALLEN e-Store

ALLEN ONLINE

## Get REAL exam practice for JEE Main 2026

with the

## Major Online Test Series!

- 13 full-syllabus tests
- 100+ additional mock tests
- 50,000+ teacher-recommended Qs. & more



Buy Now